

SYSC 5108 Exam Study: Assignment 3, Question 3

Forward propagation through a small neural network

Source question: SYSC 5108 W26 Assignment 3.pdf, Question 3

Question Summary

The assignment gives a feedforward neural network with two sensing/input neurons and three processing neurons:

inputs: neurons 1, 2, processing neurons: 3, 4, 5.

The input is

$$x_1 = 0.7, \quad x_2 = 0.3.$$

From the network figure and the submitted solution, the weights and biases are:

Neuron	bias	weight from x_1 or a_3	weight from x_2 or a_4
3	$w_{3,0} = 0.6$	$w_{3,1} = 0.1$	$w_{3,2} = 0.5$
4	$w_{4,0} = 0.8$	$w_{4,1} = 0.2$	$w_{4,2} = 0.4$
5	$w_{5,0} = 0.9$	$w_{5,3} = 0.3$	$w_{5,4} = 0.7$

The question asks for:

- the output of neuron 5 if all processing neurons use logistic activation,
- the output of neuron 5 if all processing neurons use ReLU activation,
- the matrix form for the ReLU case.

Step 1: Recall the Forward Propagation Rule

For each processing neuron, first compute the weighted sum:

$$z_j = w_{j,0} + \sum_i w_{j,i} a_i.$$

Then apply the activation function:

$$a_j = \phi(z_j).$$

Chapter 4 slides on deep feedforward networks use the same idea: each layer applies a linear transformation followed by a nonlinear activation. Goodfellow, Bengio, and Courville describe a feedforward network as a composition of layers; for a hidden layer, the standard form is a matrix multiplication plus bias followed by an activation function.

The two activation functions needed here are:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

for logistic activation, and

$$\text{ReLU}(z) = \max(0, z)$$

for rectified linear activation.

Step 2: Compute Hidden-Layer Weighted Sums

Neuron 3:

$$z_3 = w_{3,0} + w_{3,1}x_1 + w_{3,2}x_2.$$

Substitute:

$$z_3 = 0.6 + (0.1)(0.7) + (0.5)(0.3).$$

So

$$z_3 = 0.6 + 0.07 + 0.15 = 0.82.$$

Neuron 4:

$$z_4 = w_{4,0} + w_{4,1}x_1 + w_{4,2}x_2.$$

Substitute:

$$z_4 = 0.8 + (0.2)(0.7) + (0.4)(0.3).$$

So

$$z_4 = 0.8 + 0.14 + 0.12 = 1.06.$$

Part (a): Logistic Activation

With logistic activation:

$$a_3 = \sigma(z_3) = \sigma(0.82) = \frac{1}{1 + e^{-0.82}} \approx 0.694236.$$

Similarly:

$$a_4 = \sigma(z_4) = \sigma(1.06) = \frac{1}{1 + e^{-1.06}} \approx 0.742691.$$

Now compute the weighted sum for neuron 5:

$$z_5 = w_{5,0} + w_{5,3}a_3 + w_{5,4}a_4.$$

Substitute:

$$z_5 = 0.9 + (0.3)(0.694236) + (0.7)(0.742691).$$

So

$$z_5 \approx 0.9 + 0.208271 + 0.519884 = 1.628154.$$

Apply logistic activation at neuron 5:

$$a_5 = \sigma(z_5) = \sigma(1.628154) \approx 0.835917.$$

Therefore:

$a_5 \approx 0.836 \quad \text{for logistic activation.}$

Part (b): ReLU Activation

With ReLU activation:

$$a_3 = \text{ReLU}(0.82) = 0.82, \quad a_4 = \text{ReLU}(1.06) = 1.06.$$

Both values are unchanged because both weighted sums are positive.

Now compute neuron 5:

$$z_5 = 0.9 + (0.3)(0.82) + (0.7)(1.06).$$

So

$$z_5 = 0.9 + 0.246 + 0.742 = 1.888.$$

Apply ReLU:

$$a_5 = \text{ReLU}(1.888) = 1.888.$$

Therefore:

$a_5 = 1.888 \quad \text{for ReLU activation.}$

Part (c): Matrix Form for the ReLU Network

Write the input vector as

$$x = \begin{bmatrix} 0.7 \\ 0.3 \end{bmatrix}.$$

For the hidden layer containing neurons 3 and 4:

$$W^{(1)} = \begin{bmatrix} 0.1 & 0.5 \\ 0.2 & 0.4 \end{bmatrix}, \quad b^{(1)} = \begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix}.$$

Compute the hidden pre-activation vector:

$$z^{(1)} = W^{(1)}x + b^{(1)}.$$

Substitute:

$$z^{(1)} = \begin{bmatrix} 0.1 & 0.5 \\ 0.2 & 0.4 \end{bmatrix} \begin{bmatrix} 0.7 \\ 0.3 \end{bmatrix} + \begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix} = \begin{bmatrix} 0.82 \\ 1.06 \end{bmatrix}.$$

Apply ReLU elementwise:

$$a^{(1)} = \text{ReLU}(z^{(1)}) = \begin{bmatrix} 0.82 \\ 1.06 \end{bmatrix}.$$

For the output layer containing neuron 5:

$$W^{(2)} = \begin{bmatrix} 0.3 & 0.7 \end{bmatrix}, \quad b^{(2)} = 0.9.$$

Compute:

$$z^{(2)} = W^{(2)}a^{(1)} + b^{(2)}.$$

Substitute:

$$z^{(2)} = \begin{bmatrix} 0.3 & 0.7 \end{bmatrix} \begin{bmatrix} 0.82 \\ 1.06 \end{bmatrix} + 0.9 = 1.888.$$

Apply ReLU:

$$a^{(2)} = \text{ReLU}(1.888) = 1.888.$$

Step 3: Why the Logistic and ReLU Answers Differ

The logistic activation squashes all neuron outputs into the interval $(0, 1)$. Therefore the final output in part (a) is also between 0 and 1:

$$a_5 \approx 0.836.$$

ReLU does not squash positive values. It keeps positive weighted sums unchanged and only clips negative values to zero. Since every pre-activation here is positive, the ReLU network behaves like a sequence of affine transformations, giving the larger output:

$$a_5 = 1.888.$$

Concise Exam Answer

$$z_3 = 0.6 + 0.1(0.7) + 0.5(0.3) = 0.82, \quad z_4 = 0.8 + 0.2(0.7) + 0.4(0.3) = 1.06.$$

For logistic activation:

$$\begin{aligned} a_3 &= \sigma(0.82) = 0.6942, & a_4 &= \sigma(1.06) = 0.7427, \\ z_5 &= 0.9 + 0.3(0.6942) + 0.7(0.7427) = 1.6282, & a_5 &= \sigma(1.6282) = 0.8359. \end{aligned}$$

For ReLU activation:

$$\begin{aligned} a_3 &= 0.82, & a_4 &= 1.06, \\ z_5 &= 0.9 + 0.3(0.82) + 0.7(1.06) = 1.888, & a_5 &= 1.888. \end{aligned}$$

The ReLU matrix form is:

$$\begin{aligned} z^{(1)} &= \begin{bmatrix} 0.1 & 0.5 \\ 0.2 & 0.4 \end{bmatrix} \begin{bmatrix} 0.7 \\ 0.3 \end{bmatrix} + \begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix} = \begin{bmatrix} 0.82 \\ 1.06 \end{bmatrix}, \\ a^{(1)} &= \text{ReLU}(z^{(1)}) = \begin{bmatrix} 0.82 \\ 1.06 \end{bmatrix}, \\ z^{(2)} &= \begin{bmatrix} 0.3 & 0.7 \end{bmatrix} a^{(1)} + 0.9 = 1.888, & a^{(2)} &= 1.888. \end{aligned}$$

Cross References Used

- Assignment: SYSC 5108 W26 Assignment 3.pdf, Question 3.
- Local submitted Assignment 3 PDF checked for comparison, Question 3 response.
- Slides: Chapter 4 slide deck on deep feedforward networks, hidden layers, nonlinear activations, ReLU, and logistic sigmoid units.
- Textbook: Goodfellow, Bengio, and Courville, *Deep Learning*, Chapter 6, Section 6.1 on deep feedforward networks.
- Textbook: Goodfellow, Bengio, and Courville, *Deep Learning*, Chapter 6, discussion around feedforward network layers, matrix multiplication, activations, and ReLU units.
- Textbook: Goodfellow, Bengio, and Courville, *Deep Learning*, Chapter 3, Section 3.10 on the logistic sigmoid.